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Effect of refractive index mismatch on forward-to-backward ratios in SHG imaging

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Nonlinear optical imaging in the epi-direction is used to image subresolution features. We find that a refractive index mismatch between the object to be imaged and the background medium can change the far-field intensity image. As an example, we study second harmonic generation (SHG) microscopy where the forward-to-backward (F/B) ratio is used to quantify subresolution features. We show both theoretically and experimentally that the inhomogeneous refractive index in collagen tendon tissue creates near-field effects, which can change the F/B ratio by ∼20%–25%, even though the effect is negligible for most of the individual fibrils in the tissue. This is caused by the sensitivity of the backward signal on phase matching conditions. © 2018 Optical Society of America

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Epi-collected signals for coherent nonlinear imaging techniques contain valuable information. Generally, the radiation in the epi-direction is much weaker than that in the forward because the phase matching conditions promote coherent addition in the forward but destructive interference in the backward direction. However, short coherence lengths also make the epi-signal particularly sensitive to the thickness of the structure; thus, it is uniquely suited to obtain subresolution structural information. For coherent anti-Stokes Raman scattering (CARS), the epi-signal was found to contain structural information about lipid droplets on the order of 0.3λ, where λ is the anti-Stokes wavelength, while forward CARS was only sensitive to larger structures [1–3]. Epi- and forward-collected third harmonic generation can be of comparable strength for nanostructures and thin films [4–7]. The ratio of the forward-to-backward (F/B) second harmonic generation (SHG) signals (the F/B ratio) from collagen fibrils provides information about fibrils diameters down to 85 nm [8].

SHG from collagen results from the chiral structure of collagen molecules, which form highly organized hierarchical structures, enabling label-free imaging of collagen tissue [9]. The SHG signal correlates with the size and distribution of fibrils, though they are subwavelength, due to the coherent nature of SHG [10]. Properties such as fibril diameter and bundling are retrieved quantitatively from the F/B ratio through numerical modeling [11–13] and images of the F/B ratio map out of the spatial distribution of fibril size in cartilage and bone [8].

In this Letter, we find that the epi-SHG from collagen, and thus the F/B ratio, is significantly affected by the linear index of the background medium. While the effect of the refractive index is often neglected in nonlinear optical imaging of thin samples, a mismatch in refractive index between the object of interest and the surrounding medium, Δn, can have a large effect. For example, in stimulated Raman scattering and CARS imaging, detailed calculations showed that near-field enhancements can cause the perceived position of μm-sized objects to differ from their actual position by several μm in the direction along the laser axis. Further, forward-collected signals can increase by an order of magnitude [14–17].

We show through our theoretical model that a change in $\Delta n/n$—by as little as 10%—can change the measured F/B ratio in collagen tissue up to 25%. We consider mouse and rat tail tendons embedded in glycerol ($n_m = 1.47$) versus similar samples embedded in water ($n_m = 1.33$). While Δn does not significantly affect the forward signal, the epi-signal is very sensitive to the near-field enhancements that result from even such a small Δn. The altered phase matching due to these near-field enhancements significantly changes the destructive interference of the backward radiation. To test the validity of our theoretical findings, we perform experiments for thin samples of mouse tail tendon to show that changing the background medium from $n_m = 1.33$ to 1.47 alters the F/B ratio as much as 28%, consistent with our theoretical results.

We model collagen fibrils as infinitely long dielectric cylinders aligned along the y axis with a nonzero second-order nonlinear susceptibility $\chi^{(2)}$. We consider only the $\chi^{(2)}_{yyy}$ component, as it is the largest, and the fibril is illuminated by a y-polarized wave propagating in the x direction. The sign of $\chi^{(2)}_{yyy}$ depends on the fibril orientation (in either the positive or negative

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y direction), which is random [18]. The fibril has a refractive index of $n_f = 1.40$, while the background medium is either index-matched or has $n_m = 1.33$ or 1.47.

We first examine the case of a single fibril. For the index-matched case under plane wave illumination, the SHG electric field in the forward and backward directions, E_f and E_b, respectively, are given by [9]

$$E_f \propto \chi^{(2)}_{yyy} E_0^2 k^2 \pi r^2, \tag{1}$$

$$E_b \propto \chi^{(2)}_{yyy} E_0^2 k \pi r J_1(2kr) \exp(-2ikx), \tag{2}$$

where E_0 is the incident field strength of the the cylinder radius, k the SHG wave number, $J_1(2kr)$ a Bessel function of the first kind, and x the position of the fibril along the laser axis. E_f does not depend on the position of the fibril, as the forward SHG is always phase-matched to the incident light. This is true down to the level of individual scatterers within a fibril, which is why the forward field simply scales with the cylinder cross-sectional area πr^2. In the backward direction, phase matching occurs only over very short distances, which means that the fields of individual scatterers in the cylinder can interfere destructively; this occurs at the zeros of J_1. From these expressions, we obtain the F/B ratio:

$$I_f/I_b = |E_f|^2/|E_b|^2 = k^2 r^2 / J_1^2(2kr). \tag{3}$$

For SHG radiation at 405 nm, the first zero for J_1 occurs at $r = 88.2$ nm, leading to a theoretically infinite large F/B ratio.

For the index-mismatched case ($\Delta n \neq 0$), we use both an analytical and computational approach. First, we apply the Mie theory for electromagnetic scattering by a cylinder [19] to calculate the linear near-fields within the fibril. These are used to obtain the nonlinear polarization ($P \propto E^2$), which is propagated to the far-field using the free-space dyadic Green's function [20]. While this approach is computationally efficient, it does not account for the effect of Δn on the SHG scattered field. We, thus, also employ finite-difference time-domain (FDTD) that includes all effects, and the nonlinear process is implemented directly in the code [21].

In Fig. 1, we show the F/B ratio versus radius for a single fibril embedded in different background media, illuminated by a plane wave with wavelength of 810 nm. The solid black line is for $\Delta n = 0$, where we used Eq. (3). For a background medium

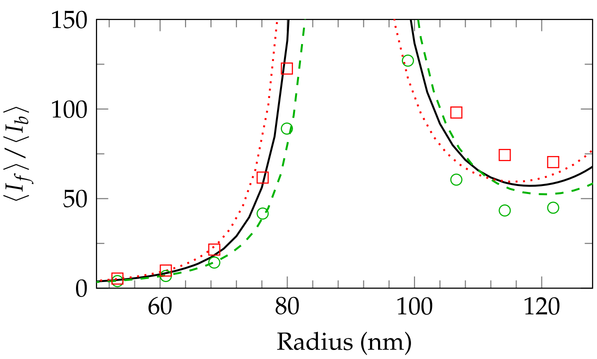

Fig. 1. F/B ratio versus radius for a single fibril embedded in $\Delta n = 0$ (black), $n_m = 1.33$ (green), and $n_m = 1.47$ (red) media, using the Mie/Green approach (dashed/dotted), FDTD (circles/squares), and Eq. (3) (black solid line).

of $n_m = 1.33$ (green) and $n_m = 1.47$ (red), we used the cylindrical Mie/Green approach (dashed lines) and FDTD (symbols). For very small radii, the F/B ratio tends to one, since fibrils act as ideal line sources with equal forward and backward radiation. For radii >70 nm, F/B ratios significantly deviate from the $\Delta n = 0$ curve. While fibrils in many biological tissues have $r < 70$ nm, there are some, such as rat and mouse tail tendons [22,23], where larger ones do occur in small numbers and, may impact the overall F/B ratio.

The difference between F/B ratios for $\Delta n = 0$ and $\Delta n \neq 0$ is partially due to the altered fields of the pump, amplified by the E^2 dependence of the induced nonlinear polarization. In Fig. 2, we plot those fields in and around a fibril with $r = 100$ nm for two different media. Regions of enhanced (yellow) and diminished (blue) field strengths up to 5% are clearly visible. However, the average field strength integrated over the fibril is only 1.3% higher than the incident light on the left ($n_m = 1.33$) and 0.9% lower on the right ($n_m = 1.47$). In the forward direction, where the SHG adds coherently, only the average field matters, and the forward field remains largely unchanged, as we verified through our calculations for Fig. 1 (not shown). However, we found that in the backward direction the destructive interference is sensitive to small changes of the field in the fibril. Near the radius where complete destructive interference takes place and the backward signal becomes small, the effect is substantial, as we see in Fig. 1.

Fibrils in tissue are always part of a bigger collection of fibrils. Even for tightly focused beams, tens to hundreds of fibrils are illuminated, each generating SHG. To get valid statistics for SHG from tendon, we generate numerically 100,000 random tendon samples by placing fibrils in a non-overlapping manner at random locations. Each fibril radius is determined at random according to the size distribution for the tissue of interest. We apply our Mie/Green approach, which is more numerically tractable than FDTD.

We consider plane wave excitation of infinitely long cylinders to enable 2D simulations, which speeds up the computation time by a factor of 300 over 3D simulations. We restrict our domain to a rectangle with a length corresponding to the Rayleigh range (4.5 μm) and a width of 1.1 μm, which is two times the beam waist. We verified this approach by comparing it to 3D Gaussian beam simulations.

The distribution of fibril radii P_r is tissue specific, as is the packing ratio ρ (the area occupied by the cylinders in the xz plane divided by the cross-sectional area of the simulation

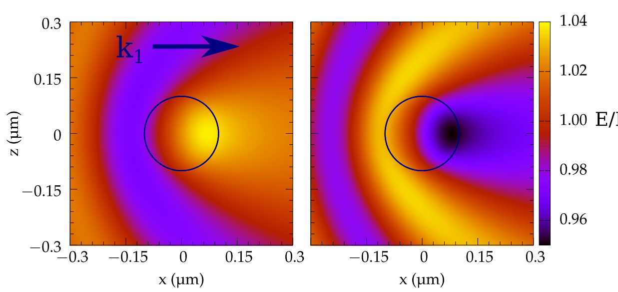

Fig. 2. Electric fields for a $\lambda_1 = 810$ nm plane wave incident from the left interacting with a dielectric cylinder calculated by cylindrical Mie theory. A cylinder of $r = 100$ nm with $n_f = 1.40$ was placed in a medium with $n_m = 1.33$ (left) and $n_m = 1.47$ (right). For $\Delta n = 0$, no distortion of the fields occur.

domain) and the positive-to-total fraction f (defined as the number of fibrils with a positive $\chi^{(2)}_{yyy}$ divided by the total number of fibrils). To numerically generate the tendon, we randomly pick a fibril's radius according to P_r, choose its position randomly inside the simulation domain, and repeat until the desired value of ρ is reached. We use a molecular dynamics simulation with reflecting boundaries to push the fibrils until they no longer overlap. The cylinders are all aligned along the y axis. This uni-directionality is typical of fibrils in tendon. Fibrils point either "up" or "down," corresponding to a positive or negative $\chi^{(2)}_{yyy}$ [18]. To obtain the F/B ratio for a collection of fibrils, the ensemble averages of the forward $\langle I_f \rangle$ and the backward $\langle I_b \rangle$ intensities are calculated separately. The F/B ratio is $\langle I_f \rangle / \langle I_b \rangle$.

We first consider the index-matched case. The SHG forward and backward fields for a single fibril, Eqs. (1) and (2), are summed analytically over all fibrils in a hypothetical sample having random radii according to P_r and random locations. This is repeated for 100,000 simulated samples and the ensemble average is derived. It is assumed for this analytical model that the fibrils are uniformly distributed along the x axis. The sign of $\chi^{(2)}_{yyy}$ is distributed with a probability f of being positive. Under these conditions, the expectation values of the forward and backward SHG are

$$\langle I_f \rangle \propto k^4(\langle r^4 \rangle N + \langle r^2 \rangle^2 N(N-1)(2f-1)^2), \tag{4}$$

$$\langle I_b \rangle \propto k^2 \langle r^2 J_1^2(2kr) \rangle N, \tag{5}$$

where N is the number of fibrils in the simulation domain, which is related to the packing fraction via $\rho = N\pi\langle r^2 \rangle / A$, where A is the cross-sectional area of the sample domain. The notation $\langle \bullet \rangle$ denotes the expectation value under $P_r(r)$.

The backward SHG does not depend on f, because the far-field SHG phase of a single fibril depends on its x coordinate [24]. As the fibril position is random, the contributions from each fibril have random phase; thus, the extra random imparted phase due to the sign of $\chi^{(2)}_{yyy}$ has no effect on the ensemble average. In the forward direction, the far-fields of each fibril add up coherently, leading to quadratic dependence on f in Eq. (4). The $(2f-1)$ in Eq. (4) is the expectation value of $\chi^{(2)}_{yyy}$ based on the probability f of the susceptibility being positive. The square of that together with $N(N-1)$ represents the average interference between the field of a fibril (of which there are N) and all its neighbors (of which there are $N-1$). The F/B ratio is independent of the number of fibrils N (and thus ρ), only if $f = 0.5$. In earlier work, it was claimed that the F/B ratio is independent of ρ [8,11]. Using Eqs. (4) and (5), we find that the independence of the F/B ratio on ρ is approximately true for rat tail tendons, where $0.47 \le f \le 0.53$ [24], but not for the instance for cartilage [25].

We now investigate the effect of Δn on the F/B ratio by using the Mie/Green approach. For collections of fibrils, this approach requires a correction to the far-field phase. The free-space Green's function uses k with $n_m = 1.40$ to get the correct phase matching conditions inside of the cylinder. However, this leads to an incorrect phase in the far-field, as the SHG travels through a medium with $n_m \neq 1.40$. This is irrelevant for a single fibril, but not for multiple fibrils, because we are adding the far-fields of each individual fibril, and thus, it is the optical distance between fibrils that matters. To correct for that, we first use the Mie/Green approach to calculate the far-field from each fibril as if it were placed at $x = 0$ μm. We then translate the fibril from position 0 to its true position x_0. If $x_0 > 0$, the input beam travels an additional distance x_0 to reach the new position, gaining an extra phase of $k_1 x_0$, where k_1 is the wave number of the input beam traveling through the background medium. Since two photons are used in the SHG process, the extra phase imparted on the SHG is doubled to $2k_1 x_0$. In the forward direction, the SHG from that fibril now has to travel a distance of x_0 less, which means an extra phase of $-k_2 x_0$. Since k_2 and $2k_1$ are effectively the same (their differences do not accumulate over the very short laser focal volume), there is no additional phase factor. However, for the backward direction, the k_2 vector is in the opposite direction, which translates into a total extra phase of $2k_2 x_0$.

We consider rat and mouse tail tendons, as they are common models used for SHG imaging that contain a small subset of larger fibrils [26]. The distribution P_r for the 13–14 week old rat tail tendon is [24]

$$P_r(r) = \frac{4r}{r_0^2} \exp\left(-\frac{2r}{r_0}\right), \tag{6}$$

where $r_0 = 65$ nm is the mean fibril radius. This particular formula was obtained by fitting the distribution measured by electron microscopy [22]. For mature mouse tail tendons, a triple normal distribution is used [23]:

$$P_r(r) = \sum_{i=1}^{3} c_i \exp\left(-\frac{(r-r_i)^2}{\sigma_i^2}\right), \tag{7}$$

where $c_1 = 10.492$, $c_2 = 17.298$, $c_3 = 15.575$, $r_1 = 18.30$ nm, $r_2 = 63.90$ nm, $r_3 = 93.65$ nm, $\sigma_1 = 8.04$ nm, $\sigma_2 = 20.56$ nm, and $\sigma_3 = 7.96$ nm. Observations of tail tendon show it is tightly packed [22], thus, we set $\rho = 0.70$.

Figure 3 shows the F/B ratio versus f for the index-matched case obtained from Eqs. (4) and (5) (orange line), and the Mie/Green approach for $n_m = 1.33$ (green pluses) and 1.47 (red crosses), using 100,000 simulated samples of rat (top) and mouse (bottom) tail tendons. A significant drop in the F/B ratio is observed for $n_m = 1.33$, while a significant increase is observed for $n_m = 1.47$. The decrease between the $n_m = 1.47$

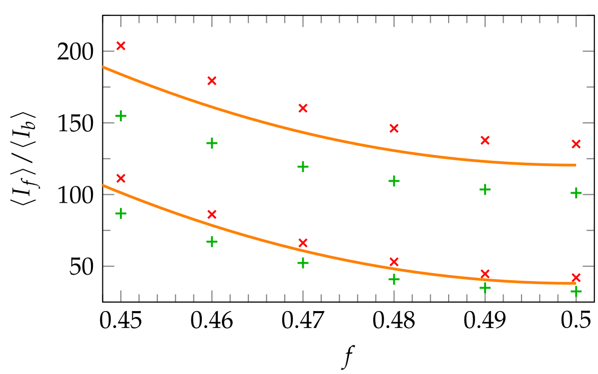

Fig. 3. Ensemble average of the F/B ratio versus f for 13–14 week old rat tail tendon (top three plots) and mature mouse tail tendon (bottom three plots) for $\Delta n = 0$ (orange line), $n_m = 1.33$ (green pluses), and $n_m = 1.47$ (red crosses).

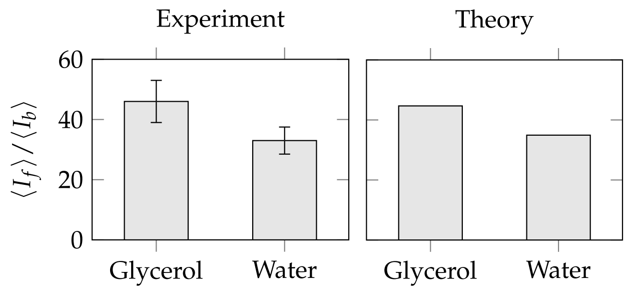

Fig. 4. Experimental F/B measurements for mature mouse tail tendons in water and glycerol (left). Theoretically predicted F/B ratio (right).

and 1.33 cases is $24.1 \pm 0.4\%$ for rat tail tendons and $21.9 \pm .4\%$ for mouse tail tendons, where these percentages are largely independent of f over the range of $f = 0.4$ to 0.5. We tested our Mie/Green approach by applying it to the index-matched case and comparing to Eqs. (4) and (5) and found perfect agreement (not shown).

To validate our theoretical conclusions, a mouse tail tendon was imaged on a custom-built laser-scanning multiphoton microscope. For a complete description of the setup, see Ref. [8]. The input laser was tuned to 810 nm and tightly focused on the sample using a water dipping lens (40×, NA 1.15, Olympus). The fibrils in the mouse tail tendon were immersed in either water ($n_m = 1.33$) or glycerol ($n_m = 1.47$) such that a refractive index mismatch exists between the fibrils and the background medium. Samples were placed on a glass slide without coverslips on top. The results after five measurements are shown in Fig. 4 (left). The decrease between the $n_m = 1.47$ and 1.33 cases is $28 \pm 9\%$ for mouse tail tendons.

The right plot shows the predicted F/B ratio for 100,000 simulated samples with $f = 0.49$, where the value of f was chosen such that the F/B ratio averaged over the water and glycerol is the same for theory and experiment. We find that the experimental results are consistent with our theoretical prediction. Note that the increase of $21.9 \pm 0.4\%$ is almost independent of f, thus, f is not a fitting parameter per se. However, the fit for f in a theory/experimental comparison reveals structural information regarding the collection of fibrils.

In this Letter, we demonstrated the near-field effects on the SHG F/B ratio for single and multiple fibrils for $\Delta n / n \approx 0.1$. These effects cannot be removed by any known experimental technique aside from finding a perfectly index-matched background; they are intrinsic to the SHG generation itself. Thus, experiments concerning the F/B ratio in tissues, where the representative fibril diameter distribution allows for fibrils larger than 70 nm, need to correct for these effects, for example, by using the theory in this Letter. This may skew quantitative subresolution diameter measurements where the F/B ratio is correlated to fibril diameter through the analytical formula valid only for the index-matched case. However, qualitative measurements such as in Ref. [8], where the fibril diameter was studied in human cartilage, are still valid because only the relative increase in diameter was important. The theory in this Letter could be applied to get more exact values for the diameter. Furthermore, the Δn effect demonstrated in this Letter also extends to other nonlinear coherent optical imaging techniques, such as epi-CARS and epi-third-harmonic generation that rely on the signal in the backward direction as the same phase matching considerations apply.

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